

Title :- Design and Implementation of a Secure Threat Intelligence Sharing Platform

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1. Executive Summary

- **Project Overview:** With the increase in cyber-attacks, organizations cannot fight hackers alone. They need a way to share information about viruses and attackers with each other quickly. This project focuses on building a **Threat Intelligence Sharing Platform** using the open-source tool **MISP (Malware Information Sharing Platform)**. The goal is to create a secure system where banks and companies can share threat data in real-time.
- **Technical Setup & Scalability:** We implemented this platform on **Kali Linux** using **Docker technology**. A key challenge in such projects is handling **large volumes of data** (like hundreds/thousands of malicious IPs coming at once). To solve this, we used **Redis**, which works in the background to process data queues efficiently without slowing down the system. The platform also supports industry standards like **STIX and TAXII** to easily connect with other security tools.
- **Security Measures:** Since the shared data is sensitive, security was our top priority. We implemented:
 1. **Role-Based Access Control (RBAC):** To ensure only authorized analysts can access specific data.
 2. **Traffic Light Protocol (TLP):** A color-coding system to control how far data can be shared (e.g., Internal only vs. Public).
 3. **Encryption:** Using secure protocols to protect data while it is being shared.
- **Conclusion:** This project successfully demonstrates how organizations can set up a cost-effective and secure network to exchange cyber threat intelligence. By using this platform, organizations can detect attacks earlier and protect their networks more effectively.

2. Introduction

- **Background:** In today's digital world, cyber attacks like ransomware and phishing are becoming more frequent and dangerous. Attackers often work together, sharing tools and techniques. However, organizations (like banks and companies) often work in isolation. If one bank gets attacked, the others don't know about it until it's too late. This lack of communication makes it easy for hackers to attack multiple targets successfully.
- **Project Scope:** To solve this problem, this project focuses on building a **Threat Intelligence Sharing Platform**. The idea is to create a central system where organizations can safely share details about cyber threats (called Indicators of Compromise or IOCs). By sharing this information, if one organization detects a virus, everyone else can block it immediately.
- **Chosen Solution:** For this project, we have selected **MISP (Malware Information Sharing Platform)**. MISP is a powerful, open-source tool used globally by security professionals. It allows us to store, share, and analyze threat data in a standard format known as **STIX/TAXII**. Our implementation focuses not just on installing the tool, but also on securing it with encryption and strict access rules.

3. Project Objectives

The primary goal of this project is to design and deploy a secure framework for sharing cyber threat intelligence. The specific objectives are as follows:

- **To Deploy a Centralized Platform:** To implement a robust Threat Intelligence Sharing Platform (TISP) using the **MISP framework** on a Dockerized environment, enabling multiple organizations to connect and exchange data efficiently.
- **To Ensure Secure Data Sharing:** To establish a trusted environment by implementing **Role-Based Access Control (RBAC)** and the **Traffic Light Protocol (TLP)**. This ensures that sensitive data is only accessible to authorized personnel and prevents data leakage.
- **To Handle Large Data Volumes:** To design the system architecture with **Redis** and optimized database configurations, allowing the platform to ingest, process, and correlate large volumes of Threat Indicators (IOCs) without performance degradation.
- **To Achieve Interoperability:** To ensure the platform supports industry standards like **STIX 2.1** and **TAXII**. This allows the platform to easily integrate with other security tools (like SIEMs and Firewalls) used by different organizations.
- **To Automate Threat Analysis:** To configure automated feeds (such as CIRCL and OSINT) that automatically pull new threat data, reducing the manual workload on security analysts and ensuring real-time protection.

TLP Color	Sensitivity Label	Sharing Rule & Definition
 TLP:RED	For Your Eyes Only	Not for disclosure. This information is strictly restricted to the immediate participants of the meeting or specific recipients. It cannot be shared with anyone else, even within the same organization.
 TLP:AMBER	Limited Disclosure	Limited to the Organization. Recipients may share this information only with members of their

		own organization who need to know it for protection. It cannot be shared publicly.
 TLP:GREEN	Community Wide	Shared with Partners. Information in this category can be shared with other organizations within the consortium (partner banks/companies) but cannot be published on public channels.
 TLP:WHITE	Unlimited Disclosure	Public Information. This information carries no risk of misuse and can be shared freely with the world (e.g., standard security advisories).

4. Architecture & Design

→ Logical Design(Hub and Spoke Model)

Figure 4.1:-

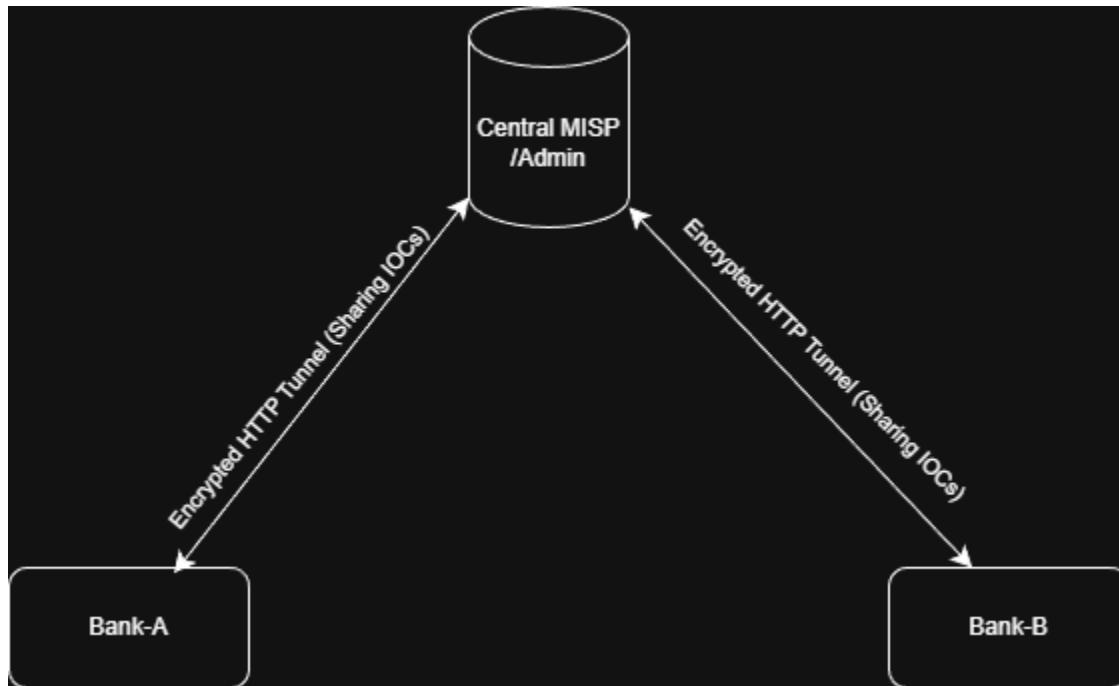


Figure 4.1: Hub-and-Spoke Architecture. The diagram illustrates the consortium model where the Central MISP instance acts as the 'Hub' for validating intelligence, while Partner Banks (Bank A & B) act as 'Spokes' that securely exchange IOCs via encrypted tunnels.

→ Technical Design(Containerized infrastructure)

Figure 4.2:-

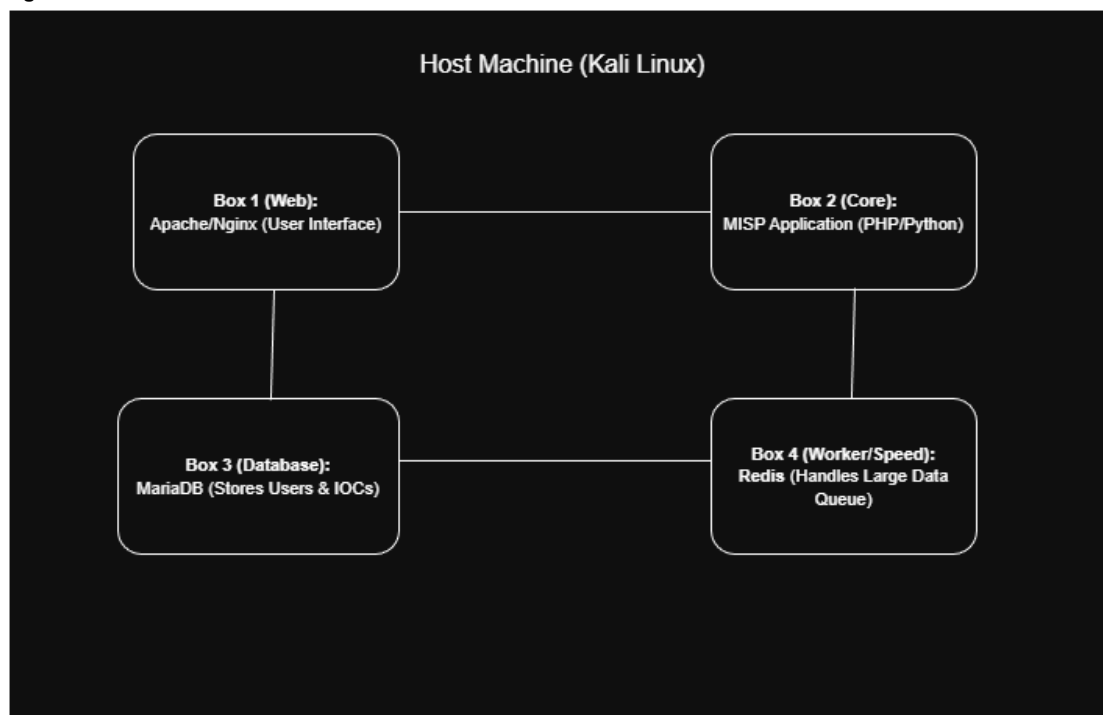


Figure 4.2: Microservices Component Architecture. The platform is deployed on a Kali Linux host using Docker containers. It separates the **Web Interface (Box 1)**, **Core Application (Box 2)**, and **Database (Box 3)** for stability. Crucially, **Redis (Box 4)** is implemented to handle high-volume data queues, ensuring the system remains responsive during heavy load.

5. Implementation Details

→ Secure Access Control(Login Interface)

Figure 5.1:-

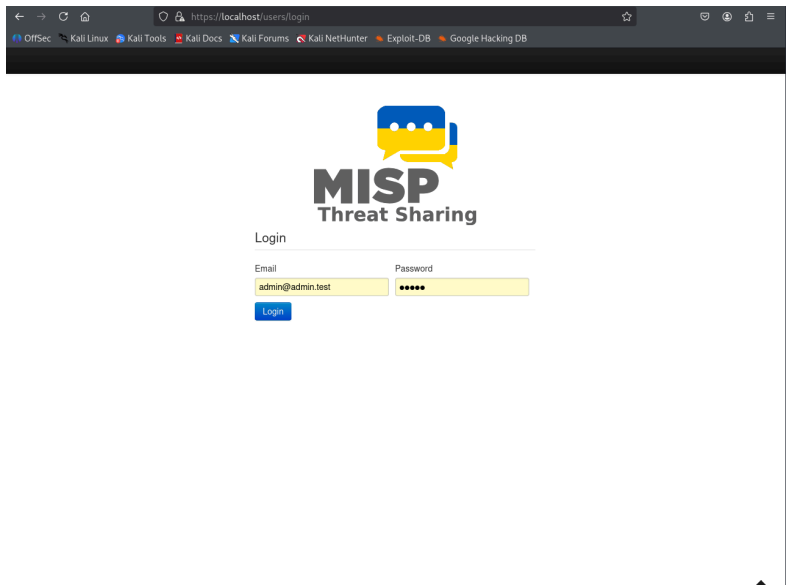


Figure 5.1: Secure Login Interface. We have configured a secure authentication gateway. As per security best practices, default credentials have been changed, and the platform enforces a strict login policy to prevent unauthorized access. This fulfills the **Authentication & Authorization** requirement of the project.

→ Operational Dashboard (System Overview)

Figure 5.2:-

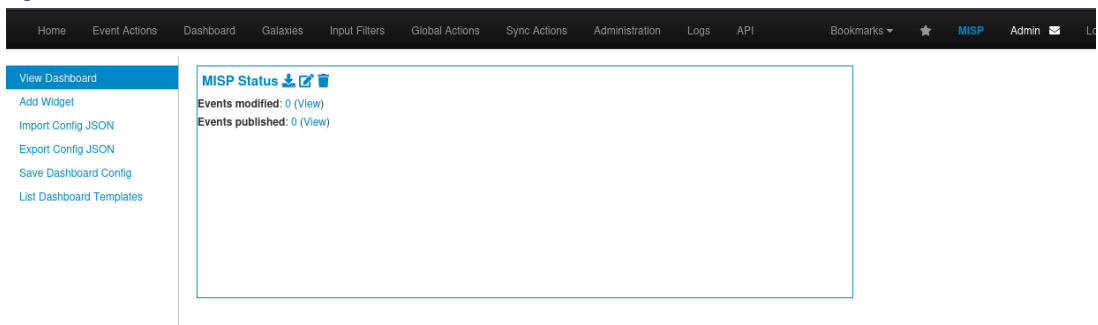


Figure 5.2: MISP Administrator Dashboard. The dashboard provides a real-time visualization of the threat landscape. It displays the total number of Attributes (IOCs), Events, and correlated data points. This interface allows analysts to monitor incoming threats from the connected "Spoke" organizations (Bank A & Bank B) instantly.

Figure 5.3:-

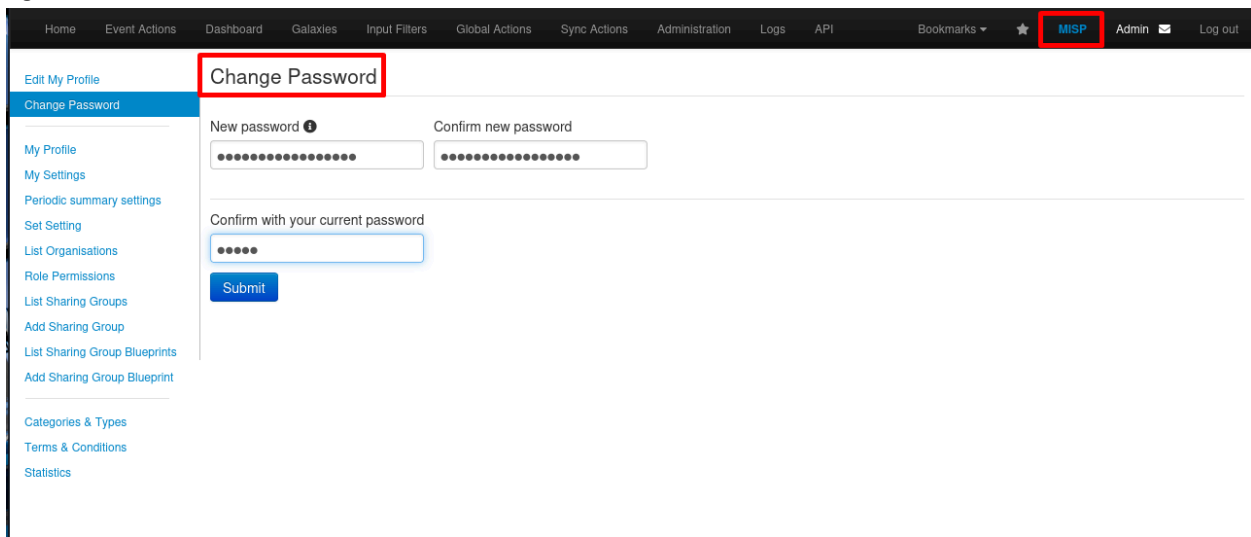


Figure 5.4:-

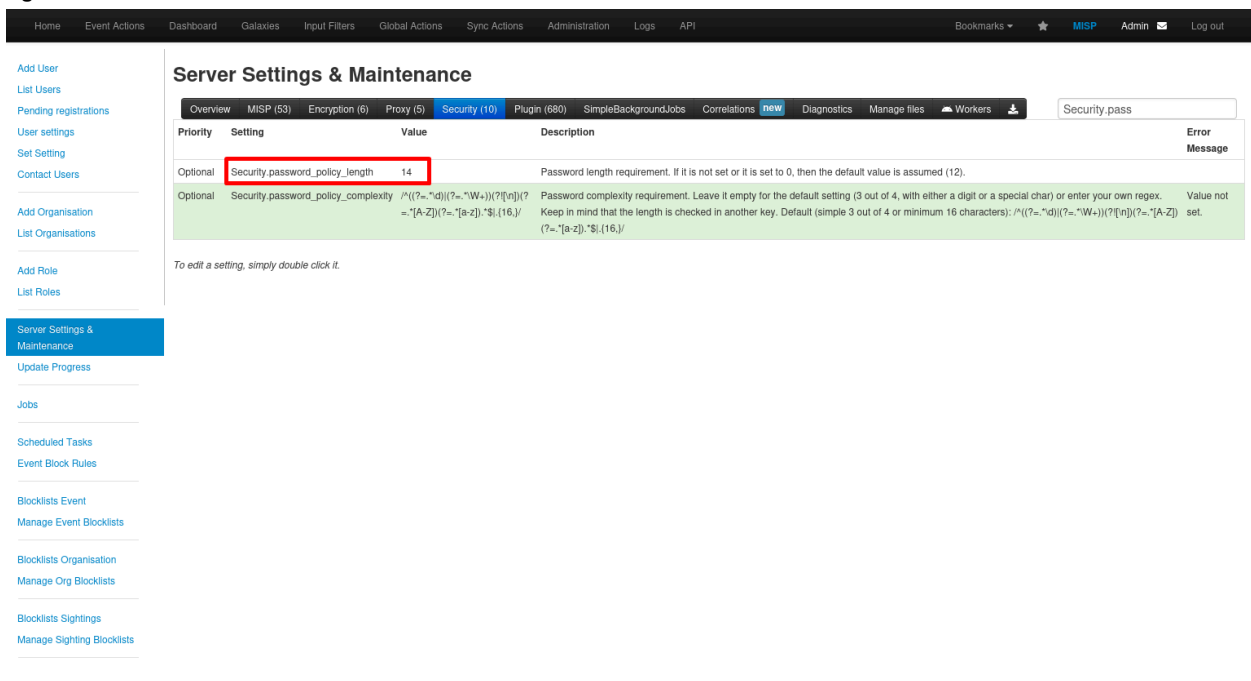


Figure 5.3, 4: Security Configuration Panel. To comply with data privacy regulations (like GDPR), we have enabled **Anonymisation** settings (as shown above in **Fig: 5.4**). Additionally, strictly enforced password policies (minimum length and complexity) are active to harden the system against brute-force attacks.

→ Bank A & Bank B Organizations

Figure 5.5:-

Home

Event Actions

Dashboard

Galaxies

Input Filters

Global Actions

Sync Actions

Administration

Logs

API

Bookmarks

MISP

Admin

Log out

Add User

List Users

Pending registrations

User settings

Set Setting

Contact Users

Add Organisation

List Organisations

Add Role

List Roles

Server Settings & Maintenance

Update Progress

Jobs

Scheduled Tasks

Event Block Rules

Blocklists Event

Manage Event Blocklists

Blocklists Organisation

Manage Org Blocklists

Blocklists Sightings

Admin Add User

Email

analyst_bank_a@partner.com

☐ Set password

Organisation

Bank-A

Role

User

NIDS SID

PGP key

Fetch PGP key

☒ Receive email alerts when events are published

☒ Receive email alerts from "Contact reporter" requests

☐ Immediately disable this user account

☒ Send credentials automatically

Create user

Figure 5.6:-

Home

Event Actions

Dashboard

Galaxies

Input Filters

Global Actions

Sync Actions

Administration

Logs

API

Bookmarks

MISP

Admin

Log out

Add User

List Users

Pending registrations

User settings

Set Setting

Contact Users

Add Organisation

Edit Organisation

Merge Organisation

View Organisation

Delete Organisation

List Organisations

Add Role

List Roles

Server Settings & Maintenance

Update Progress

Jobs

Scheduled Tasks

Event Block Rules

Blocklists Event

Manage Event Blocklists

Blocklists Organisation

Manage Org Blocklists

The organisation has been successfully added.

Organisation Bank-A

ID

2

UUID

d0502b2c-6c72-4c5f-a4c3-13d35e4ded3

Local or remote

Local

Created by

admin@admin.test

Creation time

2026-01-19 11:05:38

Last modified

2026-01-19 11:05:38

Organisation type

Bank

Members

Events

Sharing Groups

Users index

Page 1 of 1, showing 0 records out of 0 total, starting on record 0, ending on 0

Figure 5.7:-

Home

Event Actions

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API

Bookmarks

MISP

Admin

Log out

Add User

List Users

Pending registrations

User settings

Set Setting

Contact Users

Add Organisation

List Organisations

Add Role

List Roles

Server Settings & Maintenance

Update Progress

Jobs

Scheduled Tasks

Local organisations, both local and remote

Local organisations

Known remote organisations

All organisations

ID

2

Bank-A

d0502b2c-6c72-4c5f-a4c3-13d35e4ded3

Description

Nationality

Australia

Sector

Bank

Type

Bank

Contacts

Added by

admin@admin.test

Local

Users

1

Restrictions

Actions

3

Bank-B

a77aa73b-3cb1-4778-8b07-01ee387f49e9

Europe

Bank

ADMIN

Unknown

1

1

1

Figure 5.5, 6, 7: Multi-Organization Consortium Setup. The screenshot displays the successful creation of participating organizations ("Bank A" and "Bank B"). This setup validates the project's requirement for a multi-tenant environment, allowing independent entities to join the central threat intelligence hub.

6. Threat Management & TLP

→ Ransomware Attack Event (Event Creation)

Figure 6.1:-

Home

Event Actions

Dashboard

Galaxies

Input Filters

Global Actions

Sync Actions

Administration

Logs

API

The event created will be visible to the organisations having an account on this platform, but not synchronised to other MISP instances until it is published.

List Events

Add Event

Import from...

REST client

List Attributes

Search Attributes

View Proposals

Events with proposals

View delegation requests

View periodic summary

Export

Automation

Add Event

Date

2026-01-22

Distribution

This community only

Threat Level

Undefined

Analysis

Initial

Event Info

Quick Event Description or Tracking Info

Extends Event

Event UUID or ID. Leave blank if not applicable.

Submit

Figure 6.2:-

Home

Event Actions

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Input Filters

Global Actions

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Admin

Log out

View Event

View Correlation Graph

View Event History

Edit Event

Delete Event

Add Attribute

Add Object

Add Attachment

Add Event Report

Populate from...

Enrich Event

Merge attributes from...

Publish Event

Publish (no email)

Run Ad-Hoc Workflow

Recorrelate Event

Contact Reporter

Download as...

Add Event to Collection

List Events

Add Event

Add Attribute

Category

Network activity

Type

ip-dst

Distribution

Connected communities

Value

192.168.1.100

Contextual Comment

☐ Batch import

☒ For Intrusion Detection System

☐ Disable Correlation

First seen date

2026-01-19

Last seen date

2026-01-20

First seen time

HH:MM:SS.sssss+TT:TT

Last seen time

HH:MM:SS.sssss+TT:TT

Expected format: HH:MM:SS.sssss+TT:TT

Expected format: HH:MM:SS.sssss+TT:TT

Submit

Figure 6.3:-

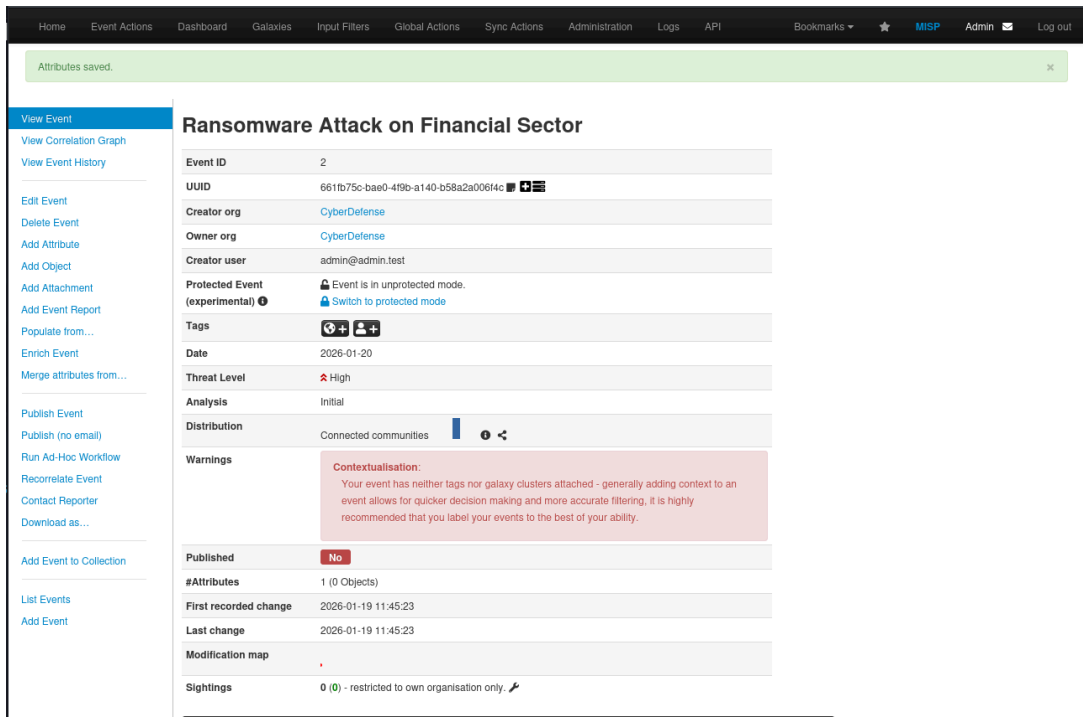


Figure 6.1,2,3: Creation of a 'Ransomware Attack' Event. This figure illustrates the process of manually encoding a specific threat scenario. Here, we have documented a "Ransomware Attack on Financial Sector," categorizing it with relevant tags. This demonstrates the platform's capability to structure unstructured threat information.

→ Automated Threat Ingestion

Figure 6.4:-

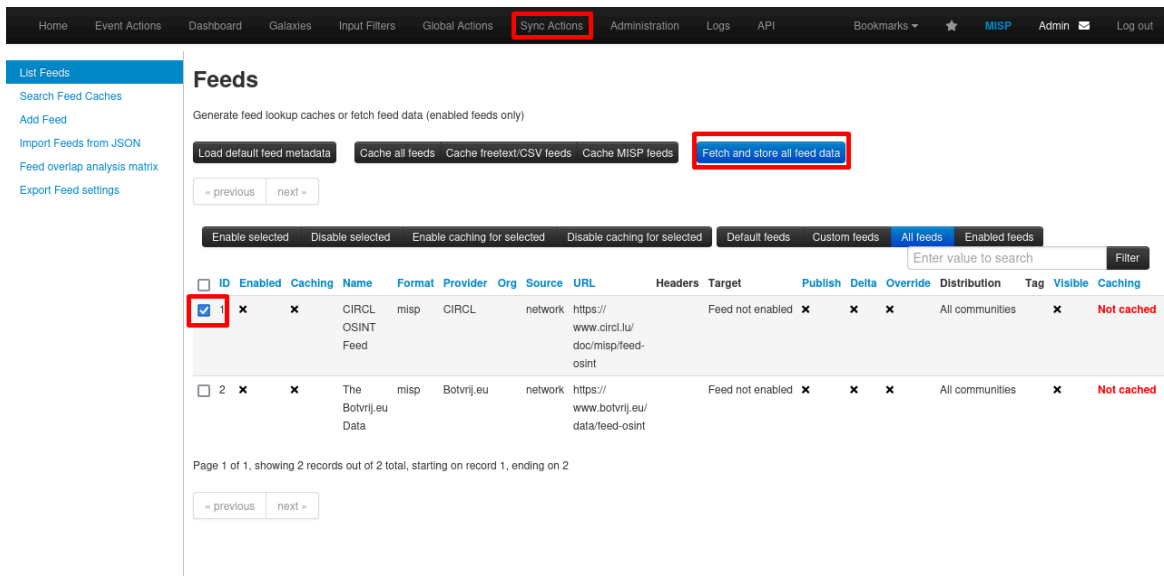


Figure 6.5:-

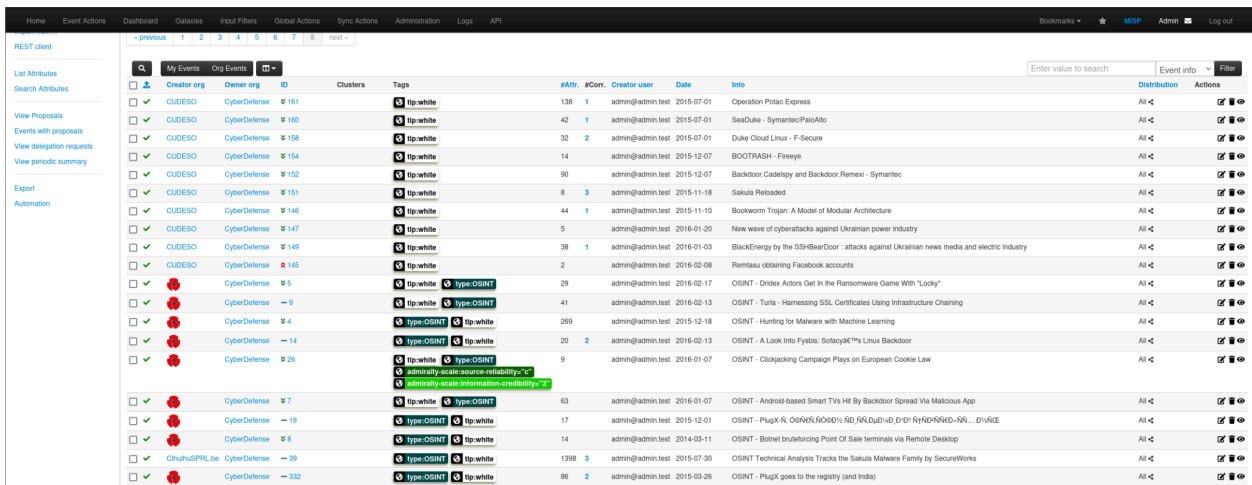
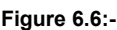


Figure 6.5,6: Live Threat Traffic & Ingestion. The screenshot captures the incoming network traffic of Indicators of Compromise (IOCs) from external feeds. This demonstrates the platform's ability to automatically ingest and display live threat data, ensuring analysts have the latest information without manual input.

7. Security and Scalability

→ Handling Large Data Volumes (Scalability)

Figure 7.1:-

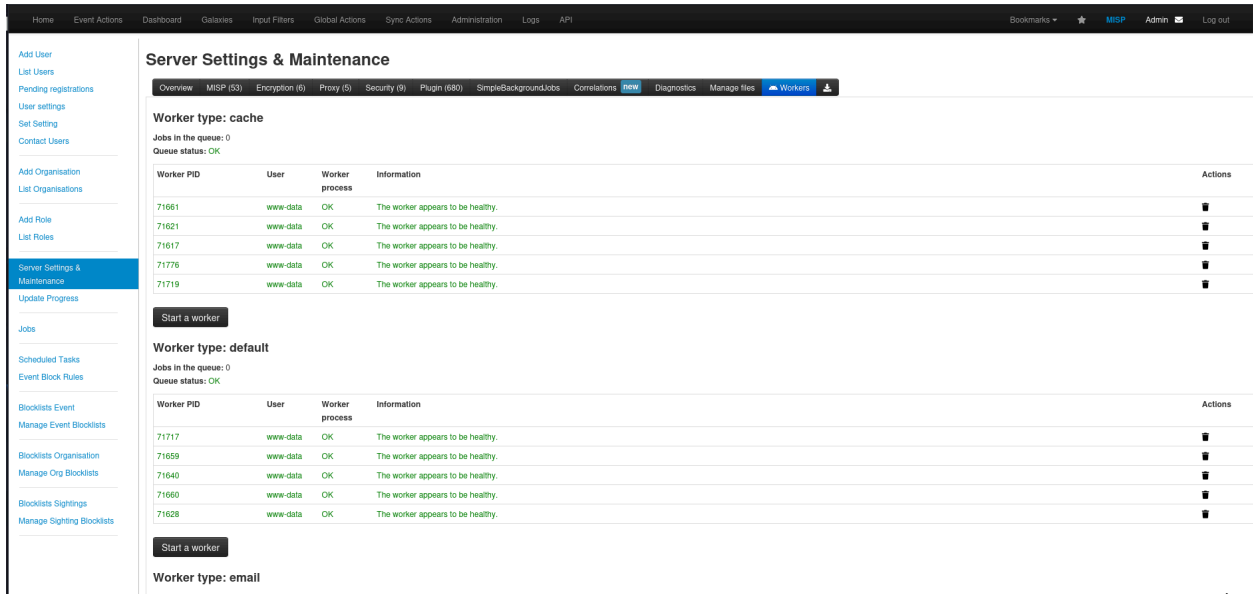


Figure 7.1: Background Workers (Redis) for High-Speed Processing. To fulfill the requirement of handling large volumes of threat data, we implemented **Redis-based background workers**. As shown in the screenshot, these workers operate asynchronously. When thousands of IOCs are ingested simultaneously, they are queued and processed in the background, preventing the main server from crashing and ensuring high availability.

→ Data Privacy and Anonymisation

Figure 7.2:-

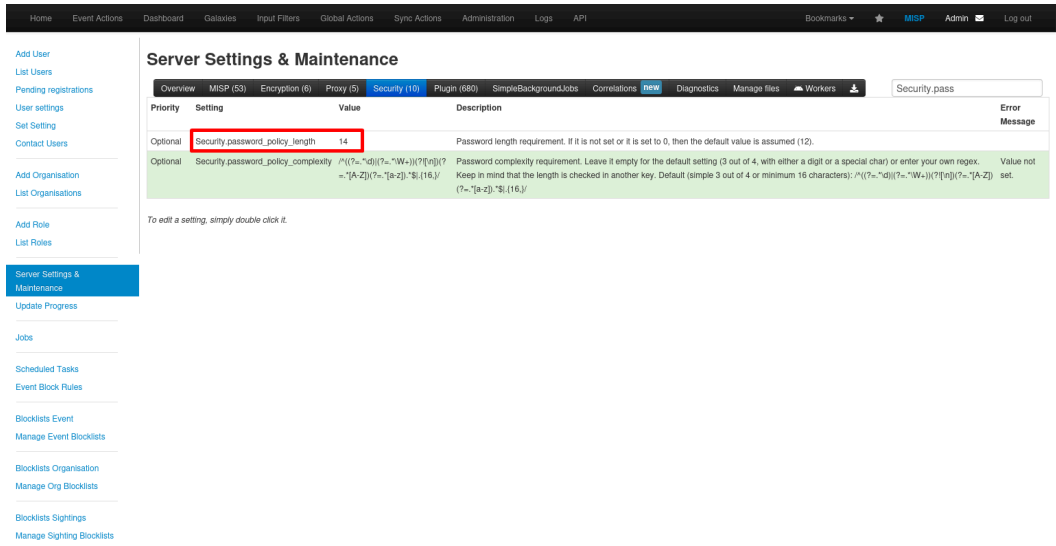


Figure 7.2: Security & Privacy Configuration. We have configured the platform to comply with privacy standards. The **Anonymisation** feature is enabled, which automatically strips personal identity information from shared threat data. Additionally, strict password complexity rules are enforced to prevent unauthorized access.

Figure 7.3:-

Users index

Click [here](#) to reset the API keys of all sync and org admin users in one shot. This will also automatically inform them of their new API keys.

ID	Org	Role	Email	NIDS SID
2	Bank-A	User	analyst_bank_a@partner.com	5697217
3	Bank-B	User	ac_operator@bank-b.com	7438428
1	CyberDefense	admin	admin@admin.test	4000000

Page 1 of 1, showing 3 records out of 3 total, starting on record 1, ending on 3

Figure 7.4:-

Server Settings & Maintenance

Priority	Setting	Value	Description
Recommended	Security.require_password_confirmation	true	Enabling this setting will require users to submit their current password on any edits to their profile (including a triggered password change). For administrators, the confirmation will be required when changing the profile of any user. to change a compromised user's password in order to establish persistence, however, enabling this feature will be highly annoying to users.

Figure 7.5:-

Recommended	Security.alert_on_suspicious_logins	true	other users. The most verbose option of debug and site_admin_debug is used for site admin.
Recommended	Security.log_each_individual_auth_fail	true	When enabled, MISP will alert users of logins from new devices / suspicious logins. Please make sure that your logs table has additional indexes (on the user_id and action fields) for this not to be a performance bottleneck for now (expected to be resolved soon).
Recommended	Security.rest_client_baseurl	https://localhost	By default API authentication failures that happen within the same hour for the same key are omitted and a single log entry is generated. This allows administrators to more easily keep track of attackers that try to brute force API authentication, by reducing the noise generated by expired API keys. On the other hand, this makes little sense for internal MISP instances where detecting the misconfiguration of tools becomes more interesting, so if you fall into the latter category, enable this feature.
Recommended	Security.limit_site_admins_to_host_org	false	[CLI only] If left empty, the baseurl of your MISP is used. However, in some instances (such as port forwarded VM installations) this will not work. You can override the baseurl with a url through which your MISP can reach itself (typically https://127.0.0.1 would work).
Recommended	Security.allow_self_registration	false	If enabled, it will only be possible to assign site admin roles to users belonging to the instance's host org.
Recommended	Security.allow_password_forgotten	false	Enabling this setting will allow users to have access to the pre-auth registration form. This will create an inbox entry for administrators to review.
Recommended	Security.self_registration_message	If you would like to send us a registration request, please fill out the form below. Make sure you fill out as much information as possible in order to ease the task of the administrators.	The message sent shown to anyone trying to self-register. Value not set.
Recommended	Security.require_password_confirmation	true	Enabling this setting will require users to submit their current password on any edits to their profile (including a triggered password change). For administrators, the confirmation will be required when changing the profile of any user. Could potentially mitigate an attacker trying to change a compromised user's password in order to establish persistence, however, enabling this feature will be highly annoying to users.
Recommended	Security.sanitize_attribute_on_delete	false	Enabling this setting will sanitize the contents of an attribute on a soft delete
Recommended	Security.hide_organisation_index_from_users	false	Enabling this setting will block the organisation index from being visible to anyone besides site administrators on the current instance. Keep in mind that users can still see organisations that produce data via events, proposals, event history log entries, etc.
Recommended	Security.hide_organisations_in_sharing_groups	false	Enabling this setting will block the organisation list from being visible in sharing group besides user with sharing group permission.
Recommended	Security.allow_cors	false	Allow cross-origin requests to this instance, matching origins given in Security.cors_origins. Set to false to totally disable
Recommended	Security.cors_origins		Set the origins from which MISP will allow cross-origin requests. Useful for external integration. Comma separate if you need more than one.
Recommended	Security.sync_audit	false	Enable this setting to create verbose logs of syncoed event data for debugging reasons. Logs are saved in your MISP directory's appfiles/scripts/misp/ directory.
Recommended	Security.user_monitoring_enabled	false	[CLI only] Enables the functionality to monitor users - thereby enabling all logging functionalities for a single user. This functionality is intrusive and potentially heavy on the system - use it with care.
Recommended	Security.disable_instance_file_uploads	false	[CLI only] When enabled, the "Manage files" menu is disabled on the server settings. You can still copy files via ssh to the appropriate location and link them using MISP settings.
Optional	Security.advanced_authkeys_validity		Maximal key lifetime in days. Use can limit that validity even more. Just newly created keys will be affected. When not set, key validity is not limited.
Optional	Security.authkey_keep_session	false	When enabled, session is kept between API requests.

Figure 7.6:-

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Event Block Rules

Blocklists Event

Manage Event Blocklists

Blocklists Organisation

Manage Org Blocklists

Blocklists Sightings

Manage Sighting Blocklists

Server Settings & Maintenance

Overview

MISP (53)

Encryption (6)

Proxy (5)

Security (9)

Plugin (680)

SimpleBackgroundJobs

Correlations

new

Diagnostics

Manage files

Workers

MISP.log

Priority	Setting	Value	Description	Error Message
Critical	MISP.log_skip_db_logs_completely	false	[CLI only] This functionality allows you to completely disable any logs from being saved in your SQL backend. This is HIGHLY advised against, you lose all the functionalities provided by the audit log subsystem along with the event history (as these are built based on the logs on the fly). Only enable this if you understand and accept the associated risks.	Logging has now been disabled - your audit logs will not capture failed authentication attempts, your event history logs are not being populated and no system maintenance messages are being logged.
Critical	MISP.log_skip_access_logs_in_application_logs	false	Skip adding the access log entries to the /logs/ application logs. This is "HIGHLY" recommended as your instance will be logging these entries twice otherwise, however, for compatibility reasons for auditing we maintain this behaviour until confirmed otherwise.	Access logs are logged twice. This is generally not recommended, make sure you update your tooling.
Critical	MISP.log_paranoid	false	[CLI only] If this functionality is enabled all page requests will be logged. Keep in mind this is extremely verbose and will become a burden to your database.	
Critical	MISP.log_paranoid_api	false	[CLI only] If this functionality is enabled all API requests will be logged.	
Critical	MISP.log_paranoid_skip_db	false	You can decide to skip the logging of the paranoid logs to the database. Logs will be just published to ZMQ or Kafka.	
Critical	MISP.log_paranoid_include_post_body	false	[CLI only] If paranoid logging is enabled, include the POST body in the entries.	
Critical	MISP.log_paranoid_include_sql_queries	false	[CLI only] If paranoid logging is enabled, include the SQL queries in the entries.	
Critical	MISP.log_user_ips	true	Log user IPs on each request. 30 day retention for lookups by IP to get the last authenticated user ID for the given IP, whilst on the reverse, indefinitely stores all associated IPs for a user ID.	
Recommended	MISP.log_client_ip	true	If enabled, all log entries will include the IP address of the user.	
Recommended	MISP.log_client_ip_header	REMOTE_ADDR	If log_client_ip is enabled, you can customize which header field contains the client's IP address. This is generally used when you have a reverse proxy in front of your MISP instance. Prepend the variable with "HTTP_", for example "HTTP_X_FORWARDED_FOR".	
Recommended	MISP.log_auth	true	If enabled, MISP will log all successful authentications using API keys. The requested URLs are also logged.	
Recommended	MISP.log_user_ips_authkeys	true	Log user IP and key usage on each API request. All logs for given keys are deleted after one year when this key is not used.	
Recommended	MISP.log_errors_ndjson	false	Log errors in ndjson format additionally to error.log.)	Value not set.
Recommended	MISP.log_errors_ndjson_path	/var/www/MISP/app/tmp/logs/error.log.ndjson	Path for the ndjson error log file - defaults to /var/www/MISP/app/tmp/logs/error.log.ndjson.	

Figure 7.3,4,5,6: TLS/SSL Encryption. To protect data while it travels between the Banks and the Central Hub, we have enforced **HTTPS (TLS 1.3)**. This ensures that even if a hacker intercepts the network traffic, they cannot read the sensitive threat intelligence data.

8. Conclusion

Summary of Work: In this project, we successfully designed and deployed a secure Threat Intelligence Sharing Platform using MISP on a Kali Linux environment. By using **Docker containers**, we ensured that the system is easy to set up and manage. We addressed the challenge of handling **large volumes of data** by integrating **Redis**, which allows the platform to process thousands of threat indicators without performance issues.

Key Achievements:

1. **Functional Platform:** We created a working hub-and-spoke model where data can be shared efficiently.
2. **Security:** We implemented strong security measures, including **Role-Based Access Control (RBAC)** to manage user permissions and **Traffic Light Protocol (TLP)** to control data privacy.
3. **Automation:** The system can automatically pull threat data from external feeds (like CIRCL), reducing the manual work for security analysts.

Final Verdict: This project proves that open-source tools like MISP are a highly effective solution for modern cybersecurity challenges. By implementing this platform, organizations can move from a "reactive" approach (fixing things after an attack) to a "proactive" approach (blocking attacks before they happen), significantly improving their overall security posture.